

Binder Remodelling with Boltz

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1. Prerequisites

1.1 Install VS Code

Download and install VS Code from the official download page:

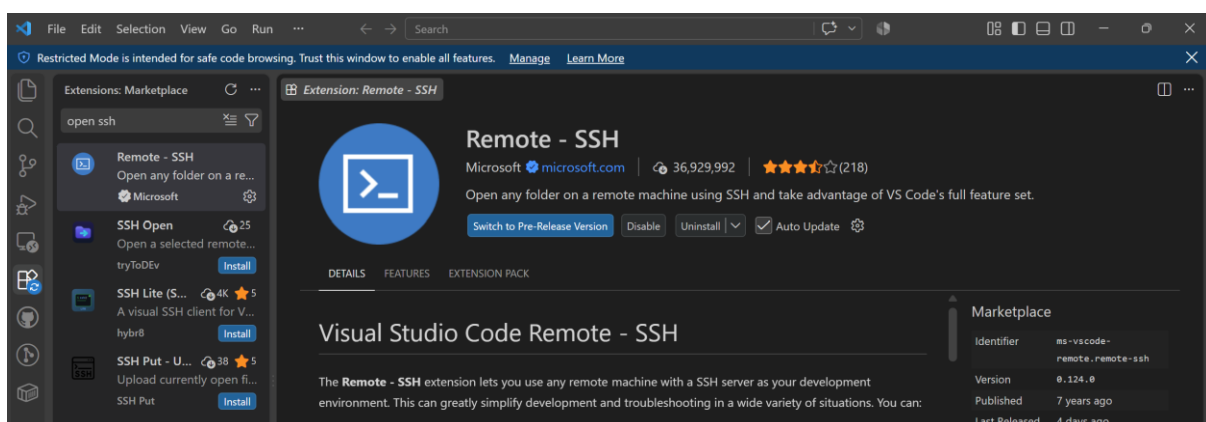
https://code.visualstudio.com/download?_exp_download=fb315fc982

1.2 Github Account

Create a GitHub account at <https://github.com/> if you do not already have one.

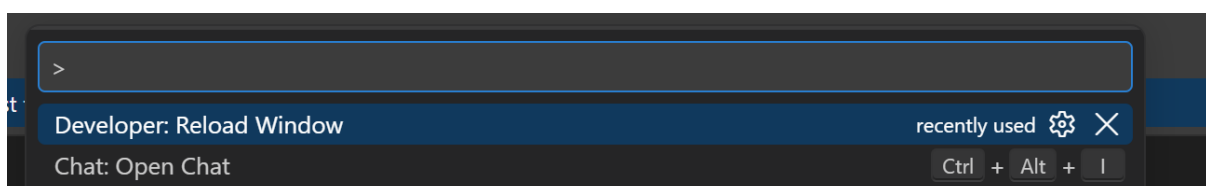
1.3 Remote- SSH extension

Install the **Remote – SSH** extension in VS Code. This extension enables you to connect to an Ewok machine from your local computer and access its computational resources directly through VS Code.



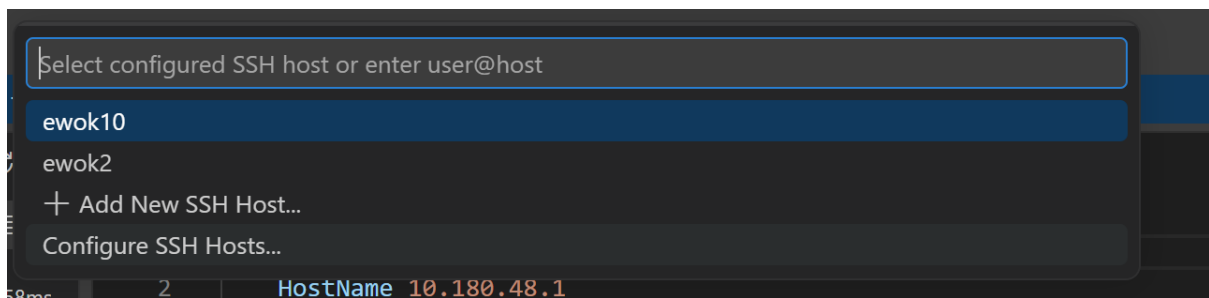
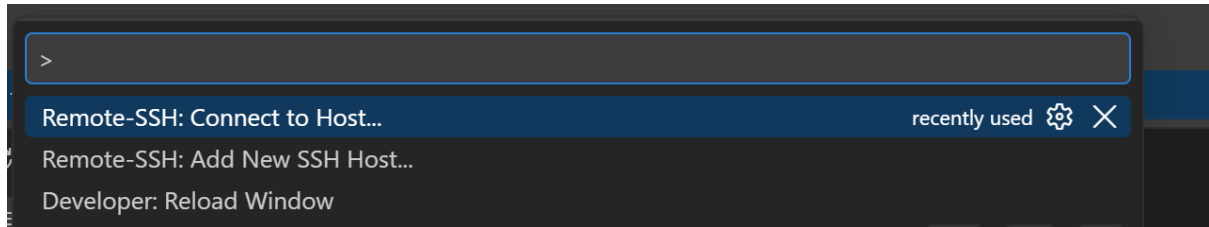
After installation, open the **Command Palette** (**Ctrl + Shift + P** on Windows or **Cmd + Shift + P** on Mac.) and type **>Developer: Reload Window**, and press Enter to ensure the extension loads correctly.

Quick tip: To run command in the **Command Palette**, make sure to start it with a **>** symbol, e.g. **>your command**.



2. Connect to the Ewok Machine

In VS Code, open the **Command Palette** and type **>Remote-SSH: Connect to Host**, then select **Configure SSH Hosts**.



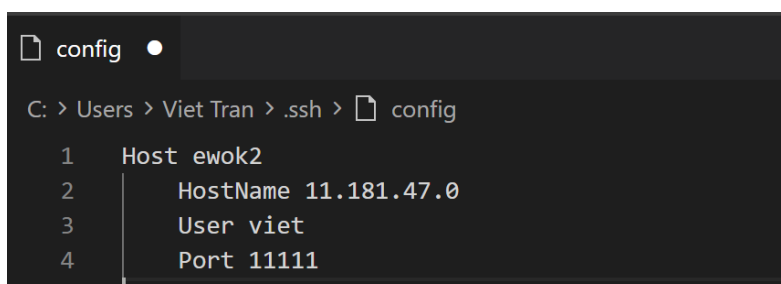
Add an entry for your Ewok machine using the following configuration format:

```
Host <ewok_machine_number>
  HostName <your_ewok_machine_ip_address>
  User <your_ewok_username>
  Port <your_ewok_port>
```

Example configuration:

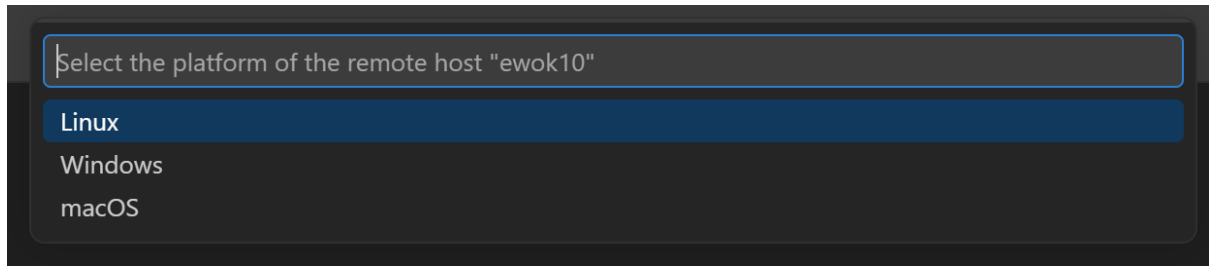
```
Host ewok2
  HostName 11.181.47.0
  User viet
  Port 11111
```

Make sure to save your changes to the config via **Ctrl + S** on Windows or **Cmd + S** on Mac.

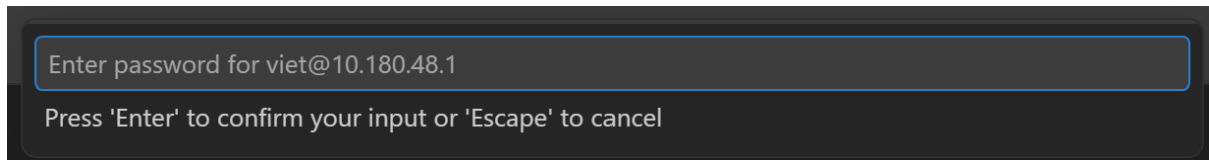


Note: Contact [Zhen He](#) or [Aaron Harris](#) for setting up and access your Ewok account.

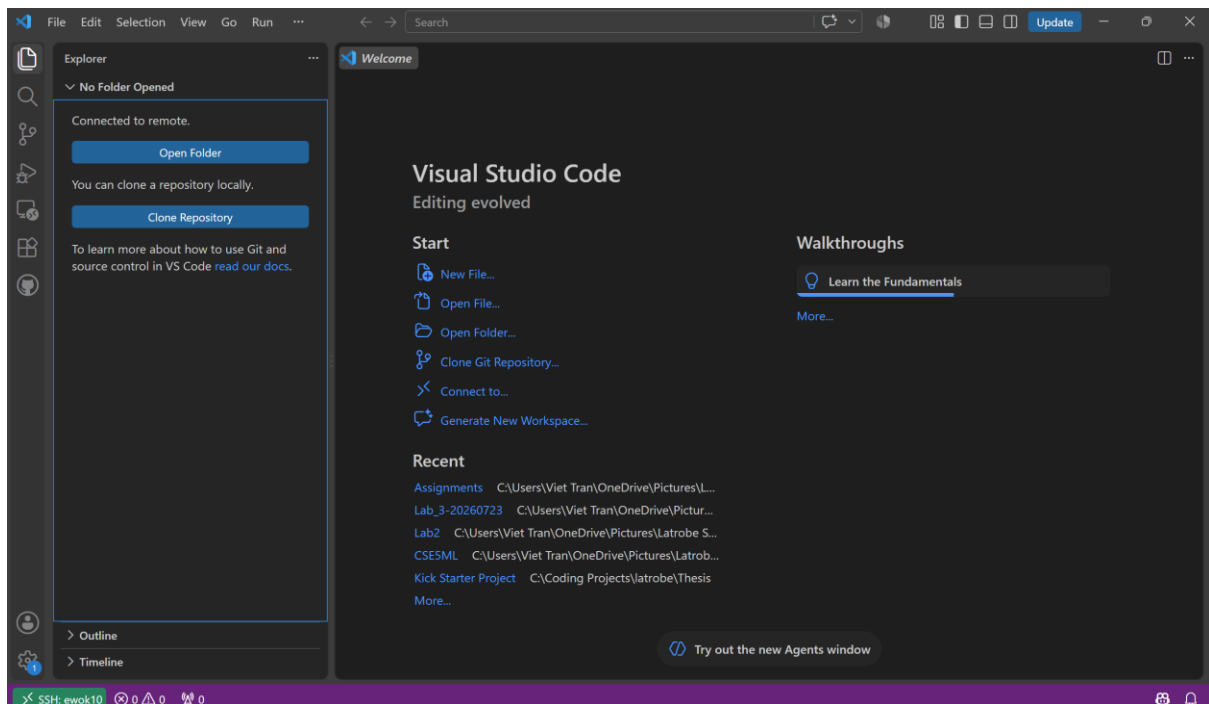
If VS Code prompts you to select the remote host's operating system, choose **Linux**



Enter your Ewok password to complete the connection.



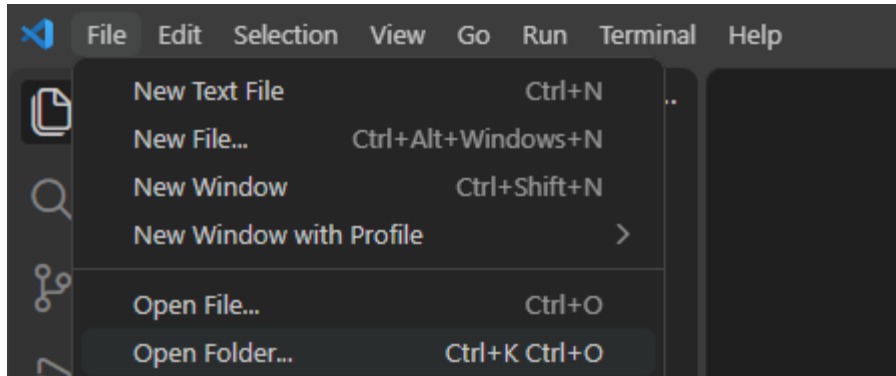
Once the connection is established, the Ewok host name should appear in the bottom-left corner of the VS Code window.



3. Clone the Boltz Repository

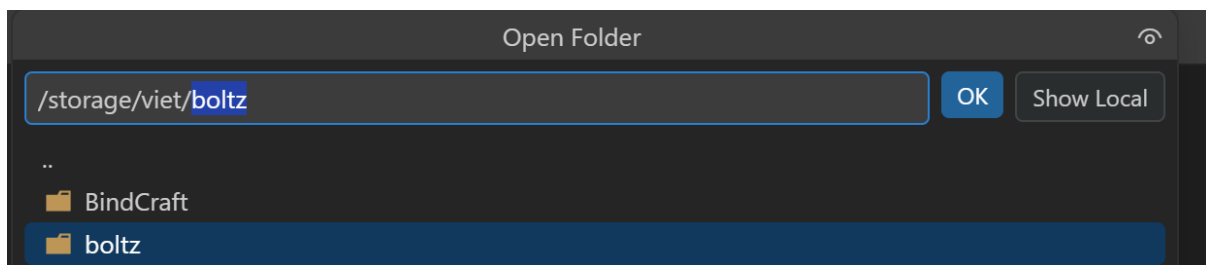
3.1 Open the Boltz Project

After connecting to the Ewok machine, select **File** → **Open Folder** in VS Code.



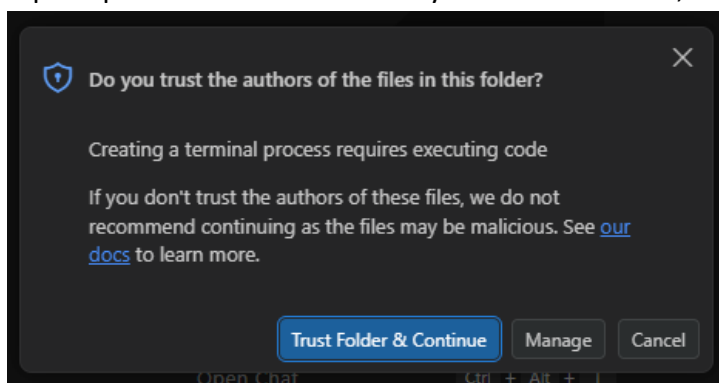
Open the directory in which Boltz is installed. Typical installation locations include:

- /home/<username>/Boltz
- /storage/<username>/Boltz



Replace <username> with your Ewok username.

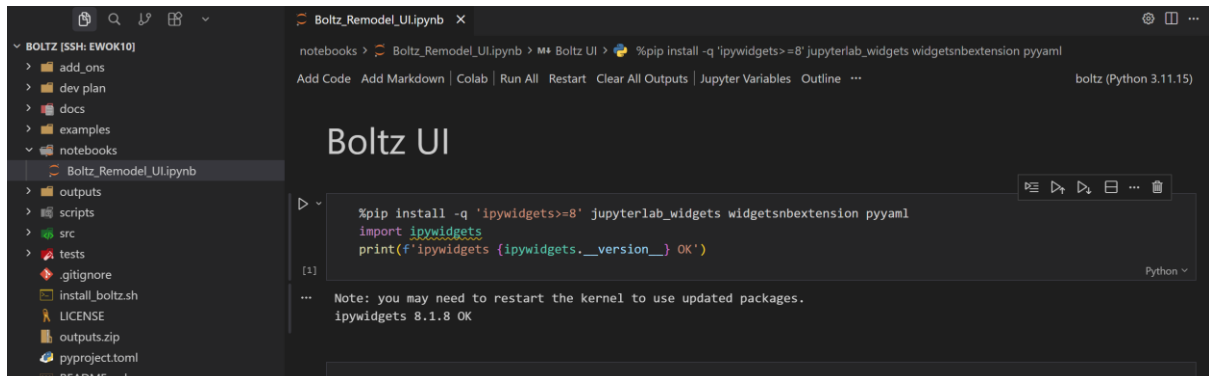
If prompted to confirm whether you trust the folder, select **Trust Folder & Continue**.



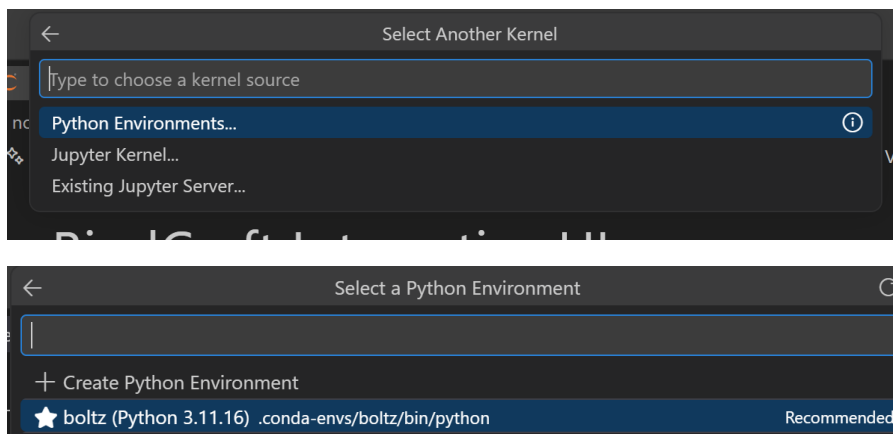
3.2 Boltz Remodel UI

3.2.1 Set Up the Notebook

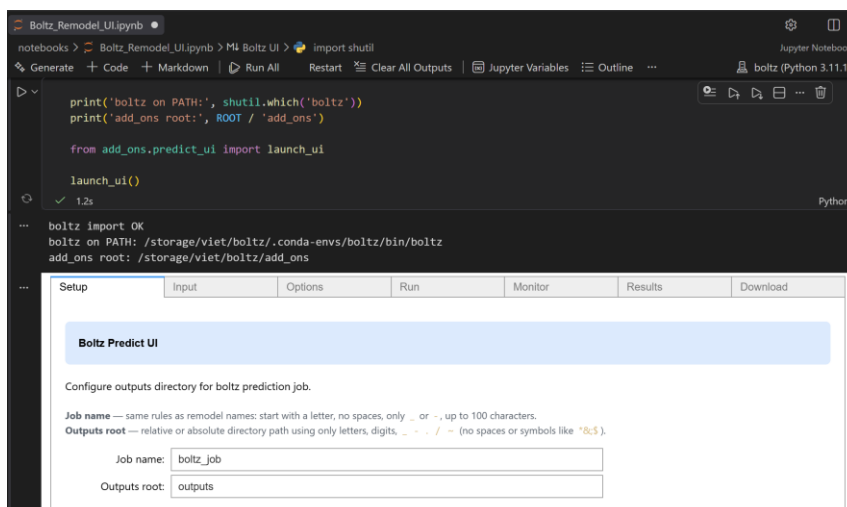
In the **Explorer** panel on the left side of VS Code, open notebooks/Boltz_Remodel_UI.ipynb



Click **Select Kernel**, choose **Python Environments**, and select the **Boltz** environment.



Then click **Run All** for dependencies installation and UI rendering.



If the UI does not appear, you may need to reload the notebook to ensure that the ipywidgets dependency is loaded correctly.

3.2.2 Run Binder Remodelling

Step 1: Configure The Remodel Job

In the **Input** tab, configure your Binder remodel job. You can either load an existing configuration or create a new one.

To load an existing configuration, select it from the **Saved config** dropdown. To create a new configuration, leave the dropdown set to **select saved config**.

Input:

Binder remodel

Edit or create a binder remodel config

Pair one target with many binders — each binder becomes its own Boltz job.

Protein sequences supported only (no DNA, RNA, or ligands).

Saved config: IFIT5_FL_bindcraft_remodel.json Refresh list

— select saved config —

IFIT5_FL_bindcraft_ewok_remodel.json

IFIT5_FL_bindcraft_remodel.json

IFIT5_FL_freebindcraft_remodel.json

IFIT5_cropped_bindcraft_ewok_remodel.json

IFIT5_cropped_bindcraft_remodel.json

IFIT5_cropped_freebindcraft_remodel.json

Job name: binder_remodel_example.json

A remodel job consists of one target and one or more binders. Each binder is processed as a separate Boltz job.

Target Definition

Field	Description
Job name	Name assigned to the Boltz job.
Target name	Name of the target protein.
Target ID	Chain ID for the target in the predicted complex, usually A.
Binder ID	Chain ID for the binder in the predicted complex, usually B.
Target sequence	Amino acid sequence of the target protein.

Binder remodel

Edit or create a binder remodel config

Pair one target with many binders — each binder becomes its own Boltz job.

Protein sequences supported only (no DNA, RNA, or ligands).

Saved config: — select saved config — Refresh list

Target

Tips for names (job, target, binder):

- For target, use something like: e.g. IFIT5_cropped or binder-1
- For binder(s), use exact name(s) from BindCraft output, e.g. IFIT5_bindcraft_ewok-Binder145_s470456_rpmnd
- Start with a letter, no spaces, only _ or - as separator, up to 100 characters

Tips for sequences:

- Paste a protein sequence (standard amino acids: ACDEFGHIKLMNPQRSTVWY)
- Letters only — numbers, spaces, and symbols are removed automatically
- Uppercased automatically: non-AA letters (e.g. S, J, O, U, X, Z) are highlighted; up to 500 residues

Job name: e.g. IFIT5_cropped_remodel

Target name: e.g. IFIT5_cropped

Target id: e.g. A Binder id: e.g. B

Target seq: Paste target protein sequence (letters A-Z only)..

Binder Definition

Each binder requires the following fields:

Field	Description
Binder name	Name assigned to the binder.
Sequence	Amino acid sequence of the binder.

Manage Binders

Enter the number of binders and click **Update binder list**. Use **Prev** and **Next** to navigate pages, with up to five binders per page.

Binders

Same naming and sequence constraints as above for every binder.

Set how many binders you need, then click **Update binder lists**. If the list is long, use Prev/Next (shows 5 at a time) or scroll.

Check binders individually, or use **Select page** for everyone on this page (selection is kept across pages). **Clear selection** unchecks all. **Delete selected** removes checked binders from the list.

When this page has fewer than 5 binders, use **Add binder** to append one without changing # binders.

binders:

← Prev

Page 1 / 1 (1 binders, 5/page)

→ Next

☒ Select page

☐ #1 of 1

Binder 1:

Sequence:

Use **Add binder** to add one (max 5 per page). To remove binders, select each via checkbox or select all on a page via **Select page**, then click **Delete selected**. Use **Clear selection** to unselect all.

Binders

Same naming and sequence constraints as above for every binder.

Set how many binders you need, then click **Update binder lists**. If the list is long, use Prev/Next (shows 5 at a time) or scroll.

Check binders individually, or use **Select page** for everyone on this page (selection is kept across pages). **Clear selection** unchecks all. **Delete selected** removes checked binders from the list. When this page has fewer than 5 binders, use **Add binder** to append one without changing # binders.

binders: **Update binder list**

← Prev Page 1 / 3 (11 binders, 5/page, 5 selected) → Next

☒ Select page ☐ Clear selection (5)

☒ #1 of 11

Binder 1:

Sequence:

☒ #2 of 11

Binder 2:

Sequence:

☒ #3 of 11

Binder 3:

Sequence:

Save Config

Finally, to apply edit to the loaded config or create new config, enter a file name and click **Save config**. Configurations are saved as JSON files in `add_ons/configs/`

Save

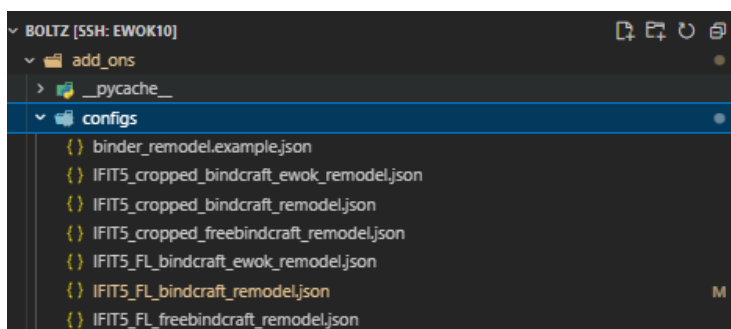
Pick a file name (same naming rules applied). Config files always saved under `add_ons/configs`.

File name:

Save config

Validation OK — saved `add_ons/configs/IFIT5_FL_bindcraft_remodel.json`

binder_remodel target=IFIT5_FL (len=482) binders=1 job_name=IFIT5_FL_bindcraft_remodel — 1 YAML file(s).



Step 2: Choose Prediction Options

In the **Options** tab, choose how Boltz should run. If you are unsure, keep the default settings.

For full details, see <https://github.com/jwohlwend/boltz/blob/main/docs/prediction.md#options>.

Prediction options

Model: Boltz-2

Device: GPU 1: NVIDIA GeForce RTX 4090 — 22909 MiB free / 24564 MiB

[Refresh GPU list](#)

Sets `CUDA_VISIBLE_DEVICES` for the Boltz job. Memory values are point-in-time snapshots; use Refresh GPU list to update them.

Samples: 5

Recycling: 10

Sampling: 200

Format: mmCIF

Seed (-1=none): -1

☒ Use MSA server (--use_msa_ser...)

☒ Use potentials (--use_potentials)

☒ Override existing (--override)

☒ Write full PAE (--write_full_pae)

Option	Description
Model	Choose Boltz-1 or Boltz-2 .
Device	Select the GPU to use. Click Refresh GPU list to update latest memory available.
Samples	Number of structure predictions generated for each binder.
Recycling	Number of refinement rounds.
Sampling	Number of calculation steps for each structure.
Format	Choose mmCIF or PDB for the output structure.
Seed	Leave as -1 unless you need repeatable results.
MSA server	Automatically prepares the sequence information Boltz needs.
Potentials	Adds extra guidance during prediction.
Override	Override existing outputs with same job name.
PAE	Saves an additional predicted alignment error file.

Step 3: Run Boltz job

When configuration and run options are ready, head to **Run** tab and click **Run predict**. Use progress bar to track selected job, and use **Abort** to stop it. Multiple jobs can be running at once, make sure to set different names for each job.

Setup

Input

Options

Run

Monitor

Results

Download

Run

▶ Run predict

Started boltz_IFIT5_FL_bindcraft_remodel (Job: IFIT5_FL_bindcraft_remodel). Change job name and Run again for parallel jobs. Attach: tmux attach -t boltz_IFIT5_FL_bindcraft_remodel

Log

```
[boltz_IFIT5_FL_bindcraft_remodel] [start] 2026-09-14 22:28:16
tmux session = boltz_IFIT5_FL_bindcraft_remodel
input = /storage/viet/boltz/outputs/IFIT5_FL_bindcraft_remodel/yaml
out = /storage/viet/boltz/outputs/IFIT5_FL_bindcraft_remodel/out
log = /storage/viet/boltz/outputs/IFIT5_FL_bindcraft_remodel/run.log
CUDA_VISIBLE_DEVICES=1
command = /home/viet/anaconda3/envs/boltz/bin/boltz predict /storage/viet/boltz/outputs/IFIT5_FL_bindcraft_remodel/yaml --out_dir /storage/viet/boltz/out
attach: tmux attach -t boltz_IFIT5_FL_bindcraft_remodel
=====
[boltz_IFIT5_FL_bindcraft_remodel] MSA server enabled: https://api.colabfold.com
MSA server authentication: no credentials provided
Checking input data.
All inputs are already processed.
Processing 0 inputs with 0 threads.

Hit [00:00, 1it/s]
^C Ctrl-C: SIGINT
```

Running jobs (parallel runs need different job names — select one to abort)

Running: IFIT5_FL_bindcraft_remodel (boltz_IFIT5_FL_bindcraft_remodel, 1m)

Refresh jobs

Abort

Job progress (selected running job)

Predicting 1/11 (9%)

Step 4: Monitor jobs

Open the **Monitor** tab to view GPU activity.

- The **nvidia-smi** panel shows available GPUs and their current usage.
- **Active Boltz jobs** lists Boltz jobs currently using GPU resources.

GPU monitor

Refresh monitor

nvidia-smi details for the current selection

How to read nvidia-smi: The report lists every GPU visible on this machine. **Memory-Usage** is current VRAM use / capacity; **GPU-Util** shows how busy the GPU is 0% means it is mostly idle, while 100% means it is very busy. This is separate from memory usage. **Pwr-Usage/Cap** is current power draw / power limit, and **Perf** is the performance state (P0 is high performance). The **Processes** section shows programs currently using GPU memory.

```
Mon Sep 14 22:33:47 2026
+-----+
| NVIDIA-SMI 580.126.09              Driver Version: 580.126.09   CUDA Version: 13.0   |
+-----+-----+-----+-----+-----+-----+-----+
| GPU  Name      Persistence-M | Bus-Id  Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf    Pwr:Usage/Cap |           |         |           | GPU-Util  Compute M. |
|=====+=====+=====+=====+=====+=====+=====+
| 0  NVIDIA GeForce RTX 4090     Off |   00000000:41:00:0  On |         | Off      |
| 34%  60C   P2             169W / 500W | 4827MiB / 24564MiB |   95%   | Default  | N/A |
+-----+-----+-----+-----+-----+-----+
| 1  NVIDIA GeForce RTX 4090     Off |   00000000:81:00:0  Off |         | Off      |
| 38%  55C   P2             312W / 500W | 15904MiB / 24564MiB |  100%   | Default  | N/A |
+-----+-----+-----+-----+-----+-----+
+-----+
| Processes: |
| GPU   GI   CI        PID   Type   Process name                  GPU Memory |
| ID    ID                                     |           | Usage      |
+-----+-----+-----+-----+-----+-----+
| 0   N/A   N/A         0        N/A   /usr/bin/python3               0MiB      |
+-----+-----+-----+-----+-----+-----+

Active Boltz jobs
```

Active Boltz jobs: This table lists running **boltz predict** processes. **GPU** is the GPU number (or - if not using VRAM yet), **PID** identifies the process, **User** is the account, **Process** is the program name, **CWD** is the working directory, and **Memory** shows VRAM use. Open this tab or press **Refresh monitor** to update.

GPU	PID	USER	PROCESS	CWD	GPU MEMORY
1	2994343	viet	boltz	/storage/viet/boltz/notebooks	54724 MiB

Step 5: Results Inspection

After the job finishes, results are saved in the outputs/ folder. Each binder has its own results folder: outputs/<job_name>/out/boltz_results_yaml/predictions/<binder_name>/

For N samples (set in **Options** tab), Boltz creates N predictions: model_0 to model_N-1. model_0 is the highest-confidence prediction.

Each sample prediction will have three main files:

File	Description
.cif or .pdb	Predicted 3D structure.
confidence_*.json	Confidence scores shown in the Results tab.
pae_*.npz	Pairwise confidence/error data.

Open the **Results** tab and select a job to view its predictions. Results are sorted by **ipTM**.

See the official <https://github.com/jwohlwend/boltz/blob/main/docs/prediction.md#output>.

Setup
Input
Options
Run
Monitor
Results
Download

Results

Pick a job folder under **outputs/** to inspect prediction metrics (sorted by ipTM).

Output folder: IFIT5_cropped_bindcraft_ewok_remodel

Loaded 55 result row(s) from /storage/viet/boltz/outputs/IFIT5_cropped_bindcraft_ewok_remodel/out/boltz_results_yaml/predictions

Results (sorted by ipTM)

design	model	ipTM	confidence_score	ptm	complex_piddt	protein_ipTM	ligand_ipTM
IFIT5_bindcraft_ewok-Binder_193_s794158_mpnn1	3	0.956	0.9332	0.9552	0.9275	0.956	0.0
IFIT5_bindcraft_ewok-Binder_193_s794158_mpnn1	2	0.9518	0.935	0.9547	0.9388	0.9518	0.0
IFIT5_bindcraft_ewok-Binder_193_s794158_mpnn1	0	0.951	0.9373	0.9537	0.9339	0.951	0.0
IFIT5_bindcraft_ewok-Binder_193_s794158_mpnn3	3	0.9475	0.9133	0.9514	0.9848	0.9475	0.0

```

outputs
├── IFIT5_cropped_bindcraft_ewok_remodel
│   ├── out / boltz_results_yaml
│   │   ├── lightning_logs
│   │   ├── msa
│   │   └── predictions
│   │       ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13
│   │       │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── confidence_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_0.cif
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_1.cif
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_2.cif
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_3.cif
│   │       │   │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_4.cif
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_0.npz
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_1.npz
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_2.npz
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_3.npz
│   │       │   │   ├── pae_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_4.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_0.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_1.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_2.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_3.npz
│   │       │   │   ├── pde_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_4.npz
│   │       │   │   ├── piddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_0.npz
│   │       │   │   ├── piddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_1.npz
│   │       │   │   ├── piddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_2.npz
│   │       │   │   ├── piddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_3.npz
│   │       │   │   ├── piddt_IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn13_model_4.npz
│   │       │   ├── IFIT5_bindcraft_ewok-Binder_172_s275174_mpnn18
│   │       │   └── IFIT5_bindcraft_ewok-Binder_176_s389396_mpnn1

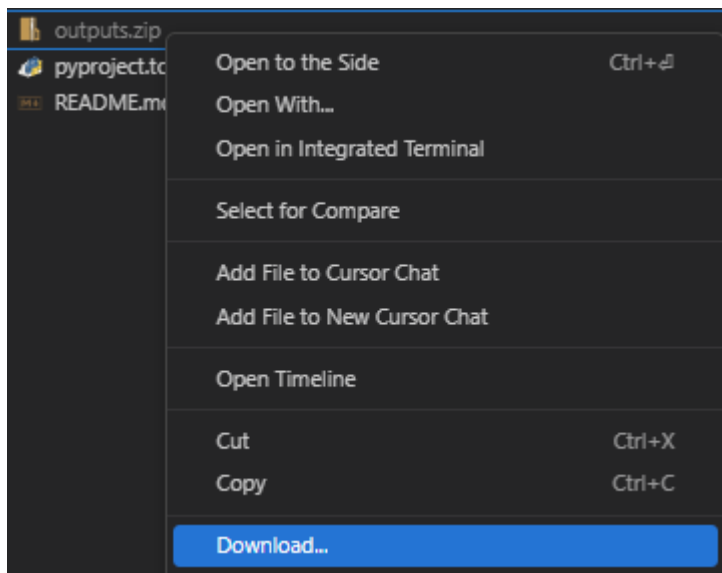
```

Step 6: Download Boltz outputs

Use the **Download** tab to create a ZIP file, especially when downloading many binders or large sequences.

Setup	Input	Options	Run	Monitor	Results	Download
Zip outputs for download Choose the entire outputs/ folder or a single design folder. The archive is written to the project root (parent of <code>notebooks/</code>); <code>/storage/viet/boltz/</code>. Zip source: Entire outputs/ folder Zip file: outputs.zip Refresh folders Create zip in project root Created <code>/storage/viet/boltz/outputs.zip</code> (569.89 MB)						

In the Explorer panel, at project root **right click** on the ZIP Boltz output then **Download**.

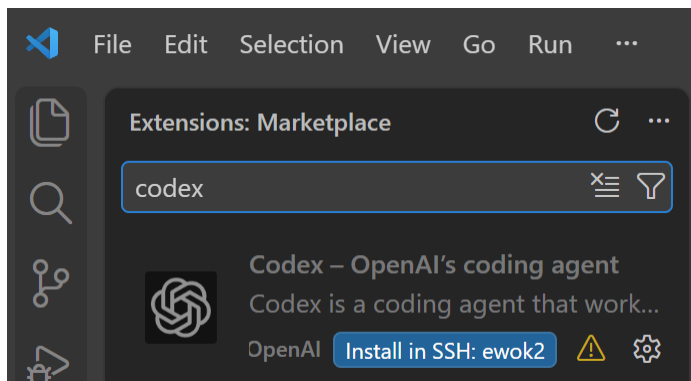


4. Useful Extensions

Codex

If you have a ChatGPT account, you can optionally install the Codex extension on the Ewok host for AI-assisted work within the project.

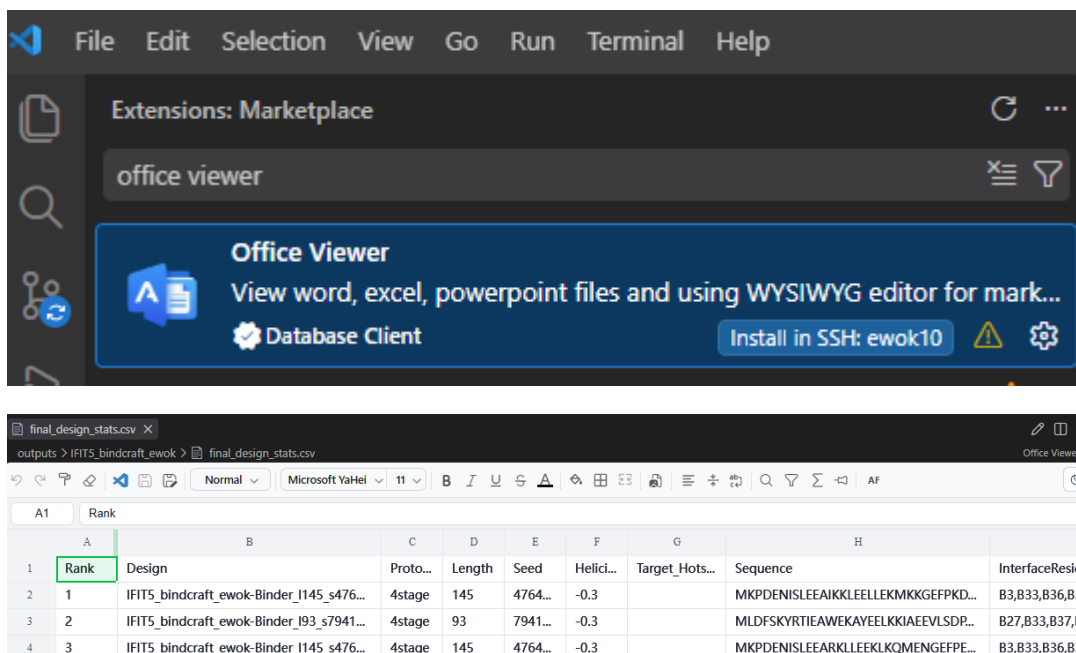
Extension: <https://marketplace.cursorapi.com/items/?itemName=openai.chatgpt>



Office Viewer

This extension helps with reading BindCraft output csv files directly in VS Code.

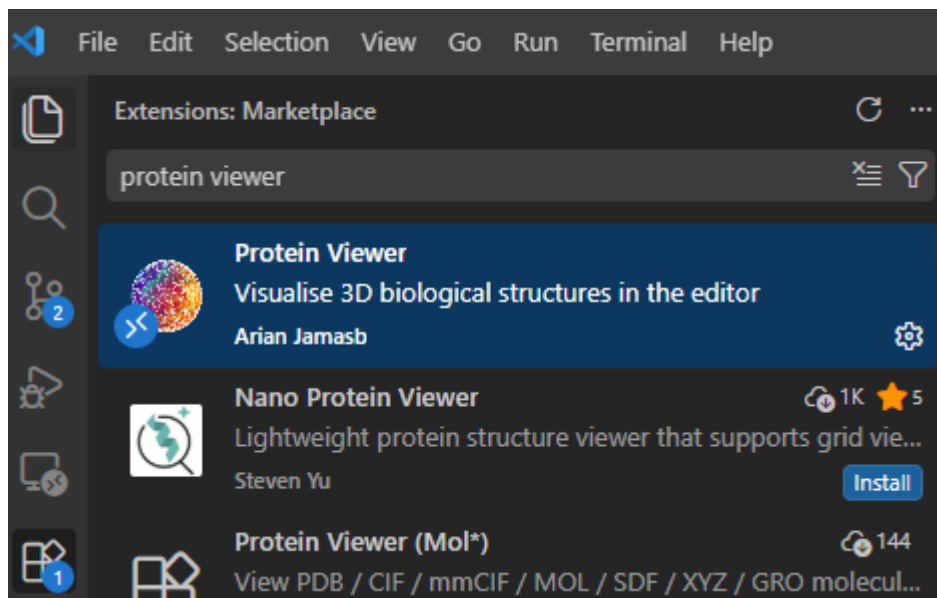
Extension link: <https://marketplace.cursorapi.com/items/?itemName=cweijan.vscode-office>



Protein Viewer

This allows inspection of target protein or generated binders PDB directly in VS Code.

Extension link: <https://marketplace.visualstudio.com/items?itemName=ArianJamasp.protein-viewer>



Right click on a PDB and select *Launch Protein Viewer* to view its 3D structure.

